

Problem 1

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Problem 1 Compute the following indefinite integrals.

(a) $\int 2t \sin(t^2) dt.$

Explanation. Let $w = t^2$. Then $dw = 2t dt$ so

$$\begin{aligned} \int 2t \sin(t^2) dt &= \int \sin(w) dw \\ &= -\cos(w) + C \\ &= -\cos(t^2) + C. \end{aligned}$$

(b) $\int \cos(x) \sqrt{\sin(x)} dx.$

Explanation. Set $v = \sin(x)$. Then $dv = \cos(x) dx$, so

$$\begin{aligned} \int \cos(x) \sqrt{\sin(x)} dx &= \int \sqrt{v} dv \\ &= \frac{2}{3} v^{3/2} + C \\ &= \frac{2}{3} (\sin(x))^{3/2} + C \end{aligned}$$

(c) $\int \left(3t^2 - 4 + \frac{1}{t} \right) e^{t^3 - 4t + \ln(t) - 9} dt.$

Explanation. This problem looks a lot more difficult than it really is.

Let $u = t^3 - 4t + \ln(t) - 9$. Then $du = \left(3t^2 - 4 + \frac{1}{t} \right) dt$, which is exactly the factor in front of the exponential.

$$\begin{aligned} \int \left(3t^2 - 4 + \frac{1}{t} \right) e^{t^3 - 4t + \ln(t) - 9} dt &= \int e^u du \\ &= e^u + C \\ &= e^{t^3 - 4t + \ln(t) - 9} + C. \end{aligned}$$